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Concordant paleolatitudes from ophiolite sequences in the northern California Coast Ranges, U.S.A.

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ABSTRACT

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Paleomagnetic data have been obtained from two ophiolite sequences in the northern California Coast Ranges: from Mount Diablo in the San Francisco Bay area and from Potter Valley, north of Clear Lake. The ophiolite exposed at Mount Diablo is part of the late Middle to Late Jurassic Coast Range ophiolite, and that exposed near Potter Valley is Late Jurassic to perhaps Early Cretaceous in age and occurs within the Franciscan assemblage. Data from the sheeted-dike complex at Mount Diablo show these rocks to be strongly overprinted, probably following uplift and erosion of the ophiolite. Samples whose primary remanent magnetization seems to be recovered yield a mean paleomagnetic pole at 30.7°N, 159.5°E with $\alpha_{95} = 5.6°$. A comparison of this pole with the Jurassic apparent polar wander path for North America indicates that the ophiolite has rotated 45° ± 7° counterclockwise relative to the craton and has not been latitudinally displaced. The diabase and pillow basalt in Potter Valley have not been strongly overprinted and data from those rocks yield a paleomagnetic pole at 79.0°N, 61.5°E with $\alpha_{95} = 6.4°$. This result indicates that the ophiolite at Potter Valley has rotated approximately 29° ± 8° clockwise, and has undergone little or no latitudinal displacement. Because of the predominantly northeastward transport of oceanic plates converging with the western margin of North America since middle Mesozoic time, the absence of appreciable northward displacement of either ophiolite fragment indicates that both formed close to the continental margin.

Introduction

It has become known in recent years that much of the North American cordillera is composed of numerous blocks of rock assemblages that are strikingly dissimilar to their adjacent neighbors. All of these blocks, or terranes, are suspect (Coney et al., 1980) in that their paleogeographic setting with respect to North America is uncertain. Some terranes are displaced pieces of the continental margin, others are fragments of oceanic crust and island arcs that were added to the continent during Mesozoic and Cenozoic times. Some of the latter may represent features that formed close to the continent, others are so distinct in character that they must be exotic to North America. Paleomagnetic studies have confirmed that many of these terranes have been transported over great distances or have undergone large rotations, or both, since their original formation. Elements of this tectonic collage within California are present in most of the physiographic provinces shown in Fig. 1.

To aid in understanding the tectonic history of this region, we conducted paleomagnetic investigations of ophiolitic rocks in two areas within the northern California Coast Ranges (Fig. 1). We are reporting the results of these investigations, even though the data are limited, because it is becoming increasingly apparent that useful paleomagnetic directions from ophiolites in the Coast Range are exceedingly difficult to obtain (see, for example, Frei and Blake, 1987; Hagstrum, 1990a). First, many of the ophiolite remnants are highly dis-

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membered, and individual elements often occur as fault-bounded slices (Hopson et al., 1981). Structural attitudes necessary to determine paleohorizontal and to correct paleomagnetic data are, therefore, often absent. Rarely is there sufficient variation in attitude to provide conclusive fold tests for stability of remanence. Second, the primary magnetization of these rocks is difficult to ascertain because even the upper parts of ophiolite sequences commonly have been metamorphosed to zeolite or greenschist facies by hydrothermal alteration at or near a spreading center (e.g., Coleman, 1977). Observable ophiolites also may have undergone additional metamorphic events related

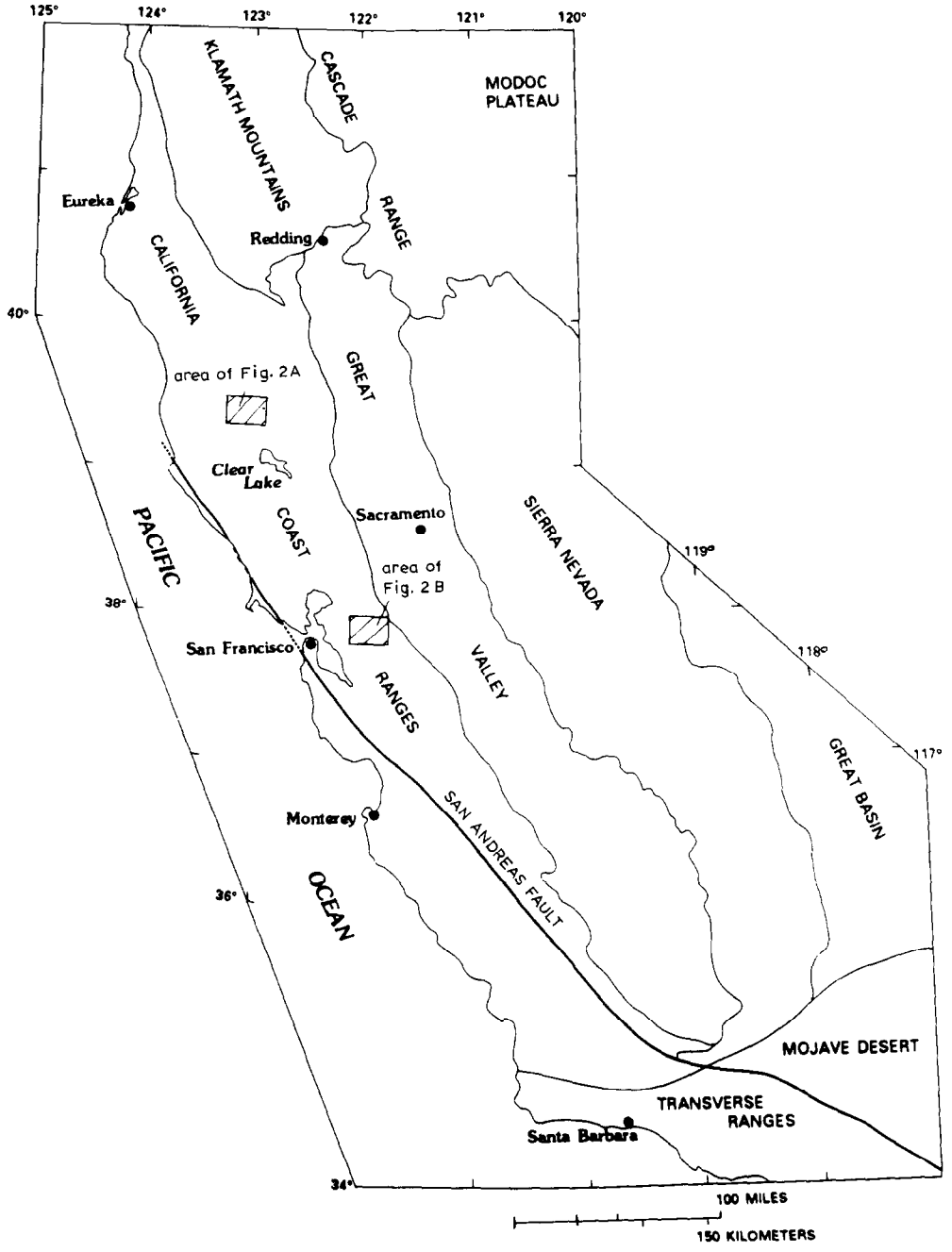


Fig. 1. Sketch showing main physiographic provinces of California and locations for Figs. 2A and 2B.

to their emplacement along continental margins or to subsequent orogenesis. Finally, many of the Coast Range ophiolite exposures are natural outcrops in which deep subaerial weathering obscures the magnetic remanence. Consequently, tectonic transport, or lack thereof, cannot be clearly resolved from an individual study and, in a later section, we review data that are available from other suspect terranes in the Coast Ranges province to see if a consistent pattern of displacements can be discerned.

Geologic setting

The geology of the California Coast Ranges is very complex because of accretion and other tectonic processes occurring along the western margin of North America. The Coast Range ophiolite is an assemblage of ultramafic, mafic, and minor felsic igneous rocks of late Mesozoic age that Bailey et al. (1970) recognized as the oceanic basement upon which Jurassic and Cretaceous strata of the Great Valley sequence were deposited. Coeval and similar ophiolite fragments that are stratigraphically below Great Valley sequence-type rocks west of the main ophiolite belt, and in the southern California Coast Ranges, also are considered to be part of the Coast Range ophiolite (Bailey et al., 1970; Hopson et al., 1981). In tectonic contact with the Coast Range ophiolite and exposed over much of the Coast Ranges are the Mesozoic and early Cenozoic rocks of the Franciscan assemblage. Deep-water sediments that are associated with ultramafic intrusive, mafic extrusive, and unusual metamorphic rocks, are typical of the Franciscan. Exposures consist of coherent rock units separated by melanges, and the assemblage is characterized by extreme variability in lithology, mineralogy, and structure (Bailey et al., 1964; Berkland et al., 1972). Although many similar rocks are collectively referred to as "Franciscan" or "Coast Range ophiolite", some occurrences may represent the amalgamation of blocks derived from different sources. Indeed, a number of rocks from each unit have been assigned to different tectonostratigraphic terranes (e.g., Silberling et al., 1984, 1987).

Our first investigation was of some ophiolitic rocks associated with the Franciscan assemblage that underlie a narrow outlier of Great Valley sequence rocks (the Middle Mountain outlier) north of Clear Lake (Figs. 1, 2A). This pile of pillow lavas, diabase/gabbro dikes, and red radiolarian chert is depositionally overlain by clastic marine sedimentary rocks and was thought to represent a seamount that was subducted beneath the Great Valley sequence during Paleocene time (Berkland, 1972a,b, 1978). This sequence of rocks is part of the Van Arsdale terrane of Silberling et al. (1984).

Our second investigation was of the diabasic rocks exposed on the northwest side of Mount Diablo east of San Francisco (Figs. 1, 2B). Mount Diablo is at the north end of the Diablo Range and is one of a number of piercement structures that are prominent along the crest of what has been termed the Mount Diablo antiform by Bailey et al. (1964). Recently, Williams (1983a) has shown that these diabasic rocks belong to a sheeted intrusive complex that is a fragment of the Middle to Late Jurassic Coast Range ophiolite. The ophiolite at Mount Diablo and other ophiolite sequences in the Diablo Range have been assigned to the Del Puerto terrane as described by Silberling et al. (1984).

A preliminary study of chert from the Van Arsdale terrane yielded radiolarians of latest Jurassic (Tithonian), or possibly earliest Cretaceous age (Berkland, 1972b), and a sample of alkali basalt stratigraphically above the chert and near the top of the volcanic section yielded a K–Ar age of about 120 Ma (Berkland, 1978). This age is increased to 122 Ma when the currently accepted K–Ar decay and abundance constants are used. However, this determination was made on a whole-rock sample from an altered basalt and should be considered to be a minimum age (R.W. Kistler, pers. commun., 1985). Chert samples obtained near the base of the ophiolite section at site 1 have yielded Late Jurassic radiolarians (B. Murchey, pers. commun., 1987), thus confirming that the ophiolite is at least as old as Late Jurassic. Whether or not parts of the ophiolite may be as young as Early Cretaceous is uncertain.

Although the diabase at Mount Diablo originally was considered to be younger than 4 Ma by Colburn (1961), Pampeyan (1963) recognized that it was much older and probably Jurassic in age. Zircons from some leucocratic dikelets, which are late-stage differentiates of the diabase screen into which the sheeted complex was intruded, have yielded an U–Pb age of 165 Ma (J.M. Mattinson, pers. commun., 1982). This age is in excellent agreement with isotopic ages ranging from 165 to 153 Ma determined for other Coast Range ophiolite localities, as summarized by Hopson et al. (1981).

Methods and results

Paleomagnetic samples were collected with a portable core drill and were oriented with a magnetic compass. Samples were measured using a cryogenic magnetometer, and alternating-field (AF) demagnetization experiments were performed using a commercial tumbling demagnetizer. Thermal demagnetization experiments on the Mount Diablo samples were performed in air using a magnetically shielded oven with an internal field less than 2 γ. Strong-field thermomagnetic measurements were made in vacuum using applied fields that ranged from 2500 to 3500 Oersteds (Oe) at heating and cooling rates of 10°C/min.

The natural remanent magnetization (NRM) direction of one specimen from each core was measured and one or more specimens from each site were subjected to progressive AF demagnetization. Although AF demagnetization proved effective for rocks from the Van Arsdale terrane, ambiguous results were initially obtained for those from Mount Diablo. For this reason, companion specimens from many of the same cores that were AF demagnetized were also subjected to thermal demagnetization in order to ascertain the most effective method of obtaining representative magnetization directions for the Mount Diablo rocks. These measurements showed that thermal demagnetization was the least effective technique and the experiments were not continued. One specimen per core from each of the sites was then progressively AF demagnetized and a stable endpoint determined for each specimen by analysis of directional changes during the demagnetization experiments and of the vector component diagrams of these data.

Van Arsdale terrane

Within the Van Arsdale terrane, only three localities were found to be sufficiently coherent and unweathered for paleomagnetic study. Ten oriented cores were obtained from diabase near the top of the volcanic section southeast of the town of Potter Valley (39.28°N, 123.04°W—site 1 in Fig. 2A), and 11 cores from pillow basalt (site 2) and 10 cores from diabase (site 3) below the Van Arsdale Reservoir dam (39.39° N, 123.11° W). The locality from which Berkland (1978) obtained the K–Ar age was found to be unsuitable for paleomagnetic sampling. Because several separate pillows were sampled at site 2, the data could represent a number of independent spot readings of the geomagnetic field, and the scatter in direction (triangles, Fig. 3A) indicates that this assumption may be true. However, it is unlikely that the full range of geomagnetic secular variation was sampled. Because of the slower cooling rate expected for intrusive rocks, some averaging of the geomagnetic field should be represented by the samples from sites 1 and 3 where the sampling was spread over the range of exposure of these intrusive bodies. Here again, however, the full range of secular variation was probably not sampled, especially at site 1 which is within the smaller of the two intrusive bodies. The most representative paleomagnetic direction for the Van Arsdale terrane can probably be obtained by combining the samples from the three localities, thereby sampling secular variation at different levels within the ophiolite. Massive diabase at site 3 is directly overlain by pillow lava at site 2, and diabase at site 1 occurs well within a sequence of pillow basalt. We caution, however, that $N = 31$ samples is likely to be an overestimate of the number of independent spot readings of the geomagnetic field, whereas $N = 3$ sites is both a gross underestimate and too small a sample to yield meaningful statistics for analysis.

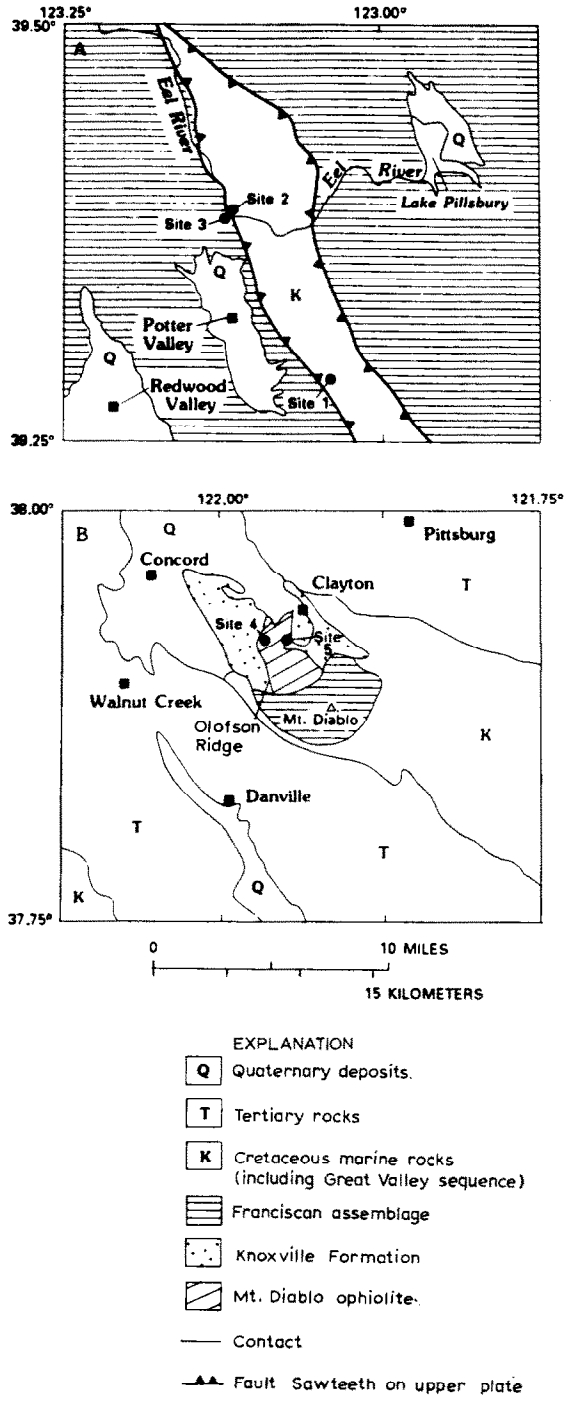


Fig. 2. Simplified geologic maps showing paleomagnetic sampling sites. Geology in the vicinity of (A) Potter Valley generalized from Jennings and Strand (1960), and (B) Mount Diablo generalized from Jennings and Burnett (1961) and Rogers (1966).

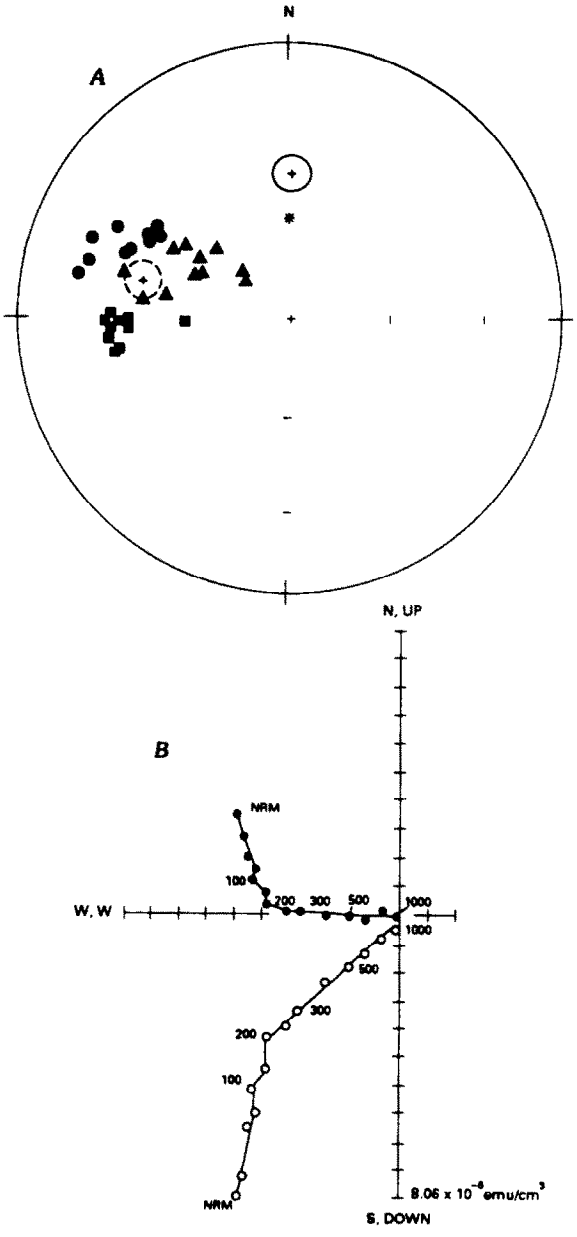


Fig. 3. (A) Equal-area projection showing average remanent magnetization directions (crosses) and 95% confidence ellipses for sites within the Van Arsdale terrane, before (dashed outline) and after (solid outline) correcting for tilt of strata. Individual specimen data (uncorrected) from site 1, squares; from site 2, triangles; and from site 3, circles. Asterisk, axial dipole field at sample locality. All directions on lower hemisphere. (B) Orthogonal projection of remanent magnetization magnitude and direction during alternating-field demagnetization for a sample from site 1. Solid symbols = projections onto the horizontal plane; open symbols = projections onto the vertical plane. Numbers represent intensity, in Oersteds, of demagnetizing field. NRM = natural remanent magnetization.

Paleomagnetic data from intrusive rocks are often difficult to interpret because such rocks commonly lack structural features necessary to

record post-emplacement tilting. In this case, however, the diabase at site 3 is directly overlain by the pillow basalts at site 2, and the attitude of the pillows can be used to correct both site means. The mean for site 1 was corrected for the attitude of the overlying Upper Cretaceous sandstone, which is in disconformable contact with the diabase and the sequence of pillow basalt that it intrudes (Berkland, 1972b, 1978). Unfortunately, structural attitudes at the three sites are nearly the same and an application of the fold test is precluded.

The magnetization of samples from each site proved to be stable, and results from all samples, after magnetic cleaning, were used in the calculation of the mean directions. Both in-situ and tilt-corrected directions are shown in Fig. 3A and listed in Table 1. A representative vector-component diagram showing the removal of remanence from one sample during AF demagnetization (Fig. 3B) indicates that a stable endpoint magnetization was readily isolated. It is apparent that the in-situ directions (Fig. 3A) do not correspond to that of any geomagnetic field known to have existed in this region since the rocks were emplaced; therefore we believe that their remanence is probably not due to a post-folding remagnetization unless subsequent rotation and/or tilting has also occurred. The change in the mean direction of each site before and after magnetic cleaning is generally away from the direction that would be produced by the present axial dipole field (ADF) at the same locality; this trend suggests that the rocks have been affected by some viscous remanent magnetization (VRM) acquired during the Brunhes Normal-Polarity Chron (i.e., the last 0.73 m.y.). Note, however, that there is little overlap of the individual specimen directions between sites (Fig. 3A) and that the data set does not have a Fisherian distribution (Fisher, 1953). These factors indicate that secular variation is not completely represented by these data, although we consider that our averaging scheme may have yielded a rough approximation of the paleofield.

Mount Diablo ophiolite

From the Mount Diablo ophiolite (Fig. 2B), 31 oriented cores were obtained from the diabase exposed in the Kaiser Sand and Gravel Company quarry (site 4), and 11 cores from diabase in the Lone Star Industries quarry (site 5) south of Clayton (37.93°N, 121.96°W). Pervasive fracturing resulting from the quarrying operations prevented us from obtaining a larger number of cores. Interpreting the results from the Mount Diablo rocks proved much more difficult than interpreting those from the Van Arsdale terrane. A total of 58 specimens was measured and the clustering of NRM directions (Fig. 4A) indicates that the remanence of these rocks was probably overprinted during the Brunhes Chron. The histogram in Fig. 4A shows that the majority of samples have directions that are within 30° of the direction that would be

TABLE 1
Paleomagnetic data from northern California ophiolites

	Site	N	Uncorrected		Corrected		R	k	α_{95}
			inc.	dec.	inc.	dec.			
Field	1, 2, 3	31	44.3	284.3	45.7	0.4	29.5284	20.4	5.9
directions	4, 5	15	7.8	119.2	−45.1	109.5	14.7348	52.8	5.3
					Corrected				
					Lat.	Long.			
Paleomagnetic	1, 2, 3	31			79.0°N	61.5°E	29.2635	17.3	6.4
poles	4, 5	15			−30.7°S	339.5°E	14.7023	47.0	5.6

Note: N = number of samples (VGP's) used in calculation of direction (pole); inc., dec. = inclination, positive downward, and declination, east of north, respectively (degrees); R = sum of N unit vectors; k = precision parameter, $(N-1)/(N-R)$; α_{95} = 95% confidence cone about mean direction and pole (degrees).

produced by the present axial dipole field at Mount Diablo. The NRM measurements also show the intensity of magnetization to be quite variable, ranging from 1.1×10^{-6} to 1.3×10^{-3} emu/cm^3 (10^{-3} emu/cm^3 = 1 A/m).

Upon demagnetization, the directions in many samples began to move away from the mean NRM direction along similar paths, a trend demonstrating that this direction was, indeed, due to overprinting. This overprint was probably acquired following uplift of the diabase and subsequent exposure to erosion and weathering, which occurred sometime after 4 Ma (Williams, 1983a).

The principal component method of Kirschvink (1980) was used to analyze the data for individual specimens whose demagnetization vector endpoints appeared to fall on great-circle paths on an equal-area projection. Bingham statistics (Onstott, 1980) were used to establish confidence ellipses for intersections of the converging remagnetization circles. Planes fitted to the vector endpoints for 14 samples converged on a direction with an inclination of 69.7° and a declination of 5.6°E, with semi-axes of the 95-% confidence ellipse equal to 18.4° and 10.3°. This direction is statistically indistinguishable from the axial dipole field direc-

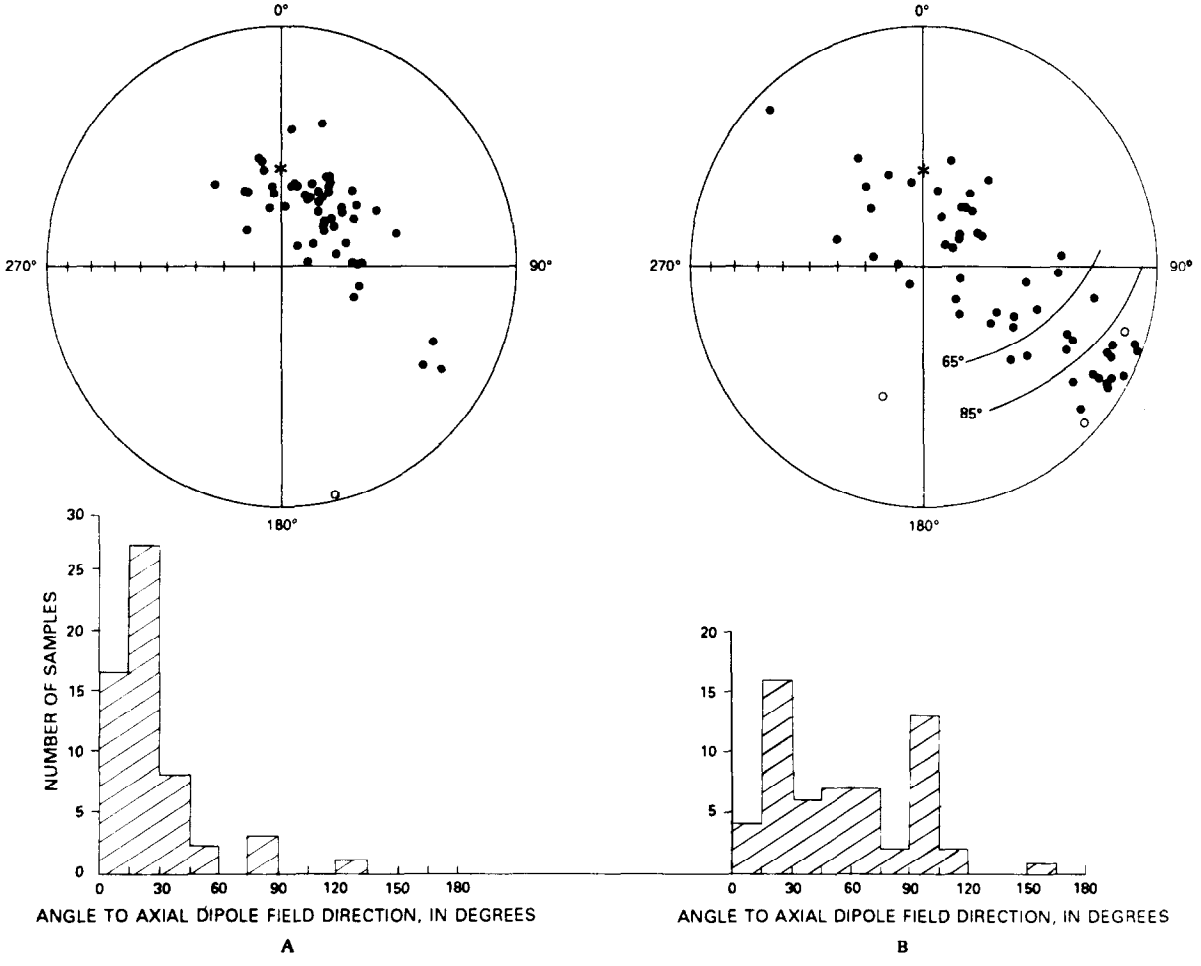


Fig. 4. Equal-area projections and histograms of average remanent magnetization directions for dike samples both (A) before and (B) after magnetic cleaning. Data not corrected for tilt. Solid symbols = directions on lower hemisphere; open symbols = directions on upper hemisphere of equal-area projection; asterisk, axial dipole field direction at sample locality. Arcs labeled *65°* and *85°* in (B) are angular distances from the axial dipole field direction.

tion at Mount Diablo and reinforces our impression of a fairly recent remagnetization of these rocks.

Unfortunately, this overprint proved to be extremely difficult to remove. Isolation of a single end-point magnetization could not be confirmed by analysis of the vector diagrams for a few of the samples (Fig. 5A). Most of the samples, however, seemed to reach a stable direction and contain only a single component of magnetization, especially at the higher demagnetization steps (Figs. 5B and 5C). A plot of these "magnetically-cleaned" directions shows a distribution that is streaked away from the ADF direction into the southeast quadrant of the equal-area projection shown in Fig. 4B. Because the overprinting field affecting our samples was less dispersed than the original field recorded by the rocks, the principal component method could not be used to determine the primary remanence in the absence of variation in structural attitude. The histogram in Fig. 4B shows that the initial concentration of directions has become broader after magnetic cleaning (out to about 75° from the ADF direction) and that a new concentration of directions has appeared approximately 100° away from the ADF direction. The histogram also shows that the break between the two concentrations of directions lies somewhere between 75° and 90° away from the ADF direction. The equal-area projection in Fig. 4B shows that two areas near this interval are devoid of data, at approximately 65° and 85° away from the ADF direction.

In order to best estimate the point at which the remanence direction is least affected by the strong remagnetization, we use the two-dimensional principal component analysis of Engebretson and Beck (1978), which provides a measure of the departure from circular symmetry of a population of unit vectors. Using this method to analyze all the data in Fig. 4B, we find that, as expected, the population is highly elongate in the direction of the ADF direction (Fig. 6A), clearly showing the effects of the overprint. We note also that the direction in one sample is close to the reversed axial dipole field (equal-area projection, Fig. 4B) and exclude it from further calculations. Because this direction is antipodal to the overprint direction, the sample was probably remagnetized during a reversed-polarity interval during the past 4 m.y., that is, subsequent to the uplift of Mount Diablo. Data that are greater than 65° away from

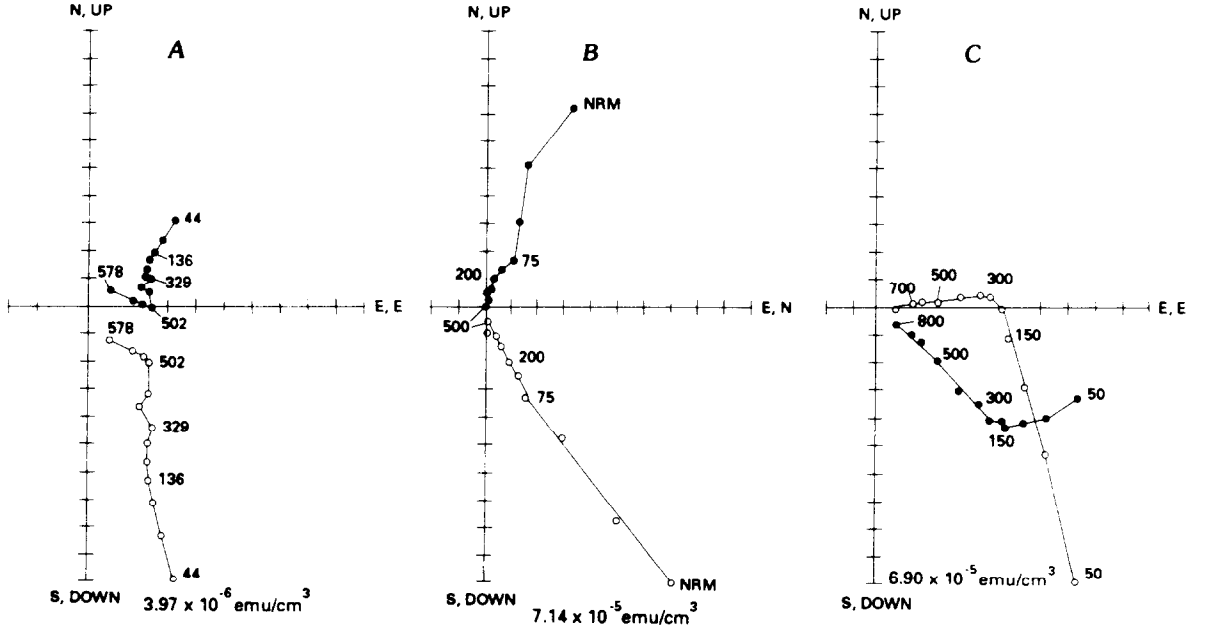
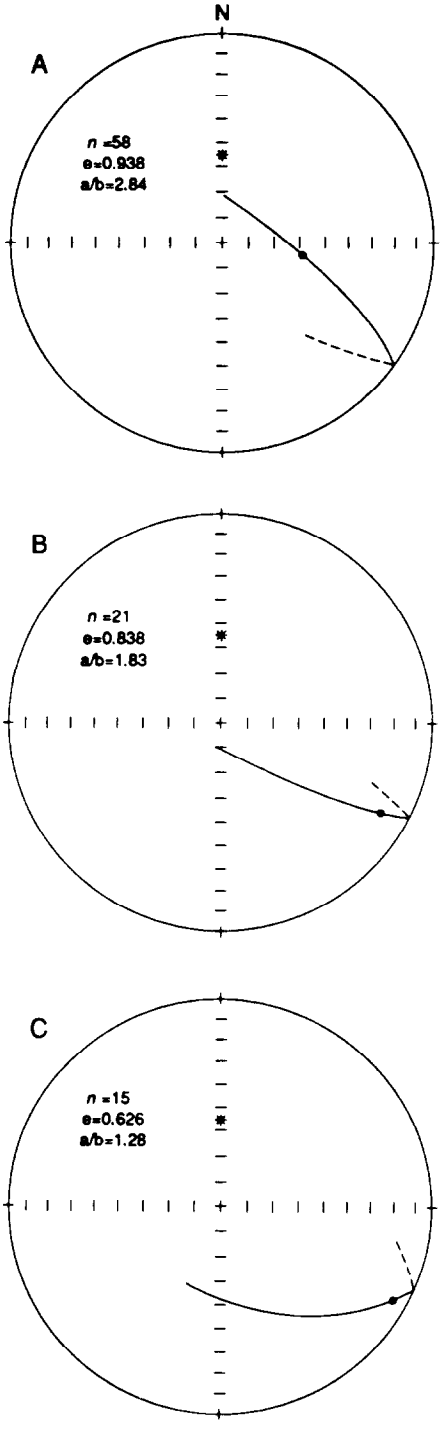


Fig. 5. Orthogonal projections (uncorrected for tilt) of remanence directions during (A) thermal and (B, C) alternating-field demagnetization. Samples shown have "magnetically-cleaned" directions that are (A) 57°, (B) 31°, and (C) 116° away from the axial dipole field direction. Numbers in (A) represent temperature, in °C, of demagnetization step. Other numbers and symbols as in Fig 3.

the ADF direction have a plane of elongation that is considerably removed from the ADF direction (Fig. 6B), and for data that are greater than 85° away, the plane of elongation is nearly transverse to the ADF direction (Fig. 6C). This last group of data is also the most circularly distributed of the three groups and is, therefore, probably the least affected by the severe overprint. We believe that this concentration of directions at about 100° away from the ADF direction best represents the stable magnetization carried by the diabase prior to the post-4 Ma overprint. Because only two of the fifteen samples in this group are from site 5, we combine the data from sites 4 and 5 to calculate a mean magnetization direction. The resulting mean direction has an inclination of 7.8° and declination of 119.2° (Table 1), which we consider to be the best estimate of the primary magnetization acquired by the sheeted intrusives at the time of their cooling.



As was the case for the ophiolite from the Van Arsdale terrane, the possibility that the effects of geomagnetic secular variation may not have been completely averaged must be considered. This is partly because our selected data set comes from only 11 individual dikes and also because the length of time required to emplace and cool individual complexes within an ophiolite sequence is not known. We will assume, however, that the directions are not grossly uncharacteristic and that they may provide a reasonable representation of the ancient geomagnetic field at this locality.

Although pillow basalt is not exposed at either quarry, it is associated with the diabase at several other localities on Mount Diablo. At two of these localities, the diabase can be seen to intrude the overlying pillow basalt, and the rock is clearly perpendicular to the bedding of the pillows. Hence, the diabase at Mount Diablo is part of a sheeted-dike complex rather than a sheeted-sill complex such as found in several other localities in the Coast Range ophiolite (Hopson et al., 1981). The

Fig. 6. Equal-area projections showing plane of elongation for magnetically cleaned samples from the Mount Diablo ophiolite. (A) Entire data set. (B) Data greater than 65°, and (C) data greater than 85° away from the axial dipole field direction. n = number of samples; e = eccentricity; a/b = major to minor axial ratio (calculated using the method of Engebretson and Beck, 1978). Partial outline of great circle (solid on lower hemisphere, dashed on upper) indicates plane of elongation of data. Asterisk = axial dipole field direction; dot = mean direction of data set (uncorrected for tilt).

only other occurrence described as a sheeted-dike complex is in the Quinto Creek ophiolite (Hopson et al., 1981), which is also in the Diablo Range located approximately 100 km southeast of Mount Diablo. Because the attitude of the dikes at Mount Diablo is similar everywhere they are exposed, striking NE–SW and dipping between 30° and 60° toward the northwest (Williams, 1983a, 1984), we use the attitude of the pillow basalt where best exposed, at Olofson Ridge approximately 3 km south of site 4 (Fig. 2B), to structurally correct the paleomagnetic data obtained from the dikes. This tilt correction restores the majority of the dikes on the main quarry face at site 4 to a near-vertical position (geometric mean of 24 measured dike orientations = 87°). The inclination and declination after correcting for the attitude of the pillows are given in Table 1.

Characteristics of the magnetization

Normal magnetic polarities are indicated for the diabase and pillow basalt at Potter Valley, and a reversed polarity for the dikes at Mount Diablo. Earlier studies (summarized by van Hinte, 1978) indicated predominantly normal polarities for Lower and Middle Jurassic rocks, and mixed polarities for Upper Jurassic rocks. More recently, however, both magnetostratigraphic studies (Ogg et al., 1984; Steiner et al., 1985, 1987) and analysis of linear magnetic anomalies on the seafloor (Handschumacher et al., 1988) seem to indicate a high reversal frequency for much of Jurassic time. The polarities recorded at the different sites are of little value, therefore, in assessing whether or not the stable remanence carried by these rocks is a primary magnetization.

Strong-field thermomagnetic measurements of four samples from the Mount Diablo dikes and single samples from each of the sites in the Van Arsdale terrane show Curie temperatures (T_c) ranging from 545° to 580°C. Curie temperatures between 560° and 580°C are considered to be typical of basaltic rocks from ophiolites that have undergone greenschist-facies metamorphism (Banerjee, 1980; Beske-Diehl and Banerjee, 1980) because the primary magnetic phase in extrusive (and perhaps high-level intrusive) rocks is titanomagnetite with an ulvospinel content about 60 mol. % and Curie temperatures between 120° and 150°C (e.g., Johnson and Hall, 1978). This primary titanomagnetite is easily oxidized at low temperatures to titanomaghemite with Curie temperatures increasing continuously to about 400°C. Inversion of metastable titanomaghemite to low-Ti titanomagnetite ($T_c \simeq 580$°C) and ilmenite ($T_c \simeq -215$°C) seems to occur near the zeolite/greenschist-facies boundary (Beske-Diehl and Banerjee,

TABLE 2
Magnetic properties of northern California ophiolites

Site	N	Property	Value	Range of 1 std. dev.
1	10	J_{nrm}	1.63×10^{-5}	6.27–0.42
		χ	9.22×10^{-5}	16.9 –5.02
		Q	0.35	0.73–0.16
2	11	J_{nrm}	1.65×10^{-4}	4.24–0.64
		χ	4.44×10^{-4}	12.0 –1.65
		Q	0.66	1.04–0.42
3	10	J_{nrm}	9.54×10^{-4}	52.9 –1.72
		χ	1.06×10^{-3}	3.11–0.36
		Q	1.76	3.58–0.87
4, 5	42	J_{nrm}	1.53×10^{-4}	8.47–0.28
		χ	9.66×10^{-4}	39.4 –2.37
		Q	0.31	0.60–0.16

Note: N = number of samples; J_{nrm} = intensity of natural remanent magnetization (emu/cm^3); χ = volume susceptibility (cgs); Q = Koenigsberger ratio ($J_{nrm}/\chi F$, where $F = 0.51$ Oe). All values are geometric means.

1980). Petrographic analyses by Williams (1983a) do show that the Mount Diablo sheeted dike complex and some of the extrusive basalt have been metamorphosed to the greenschist facies. However, a Curie temperature of 545°C in one of the Mount Diablo dikes suggests that the low-Ti titanomagnetite at that locality could be primary and not a result of the disproportionation of titanomaghemite.

The greenschist-facies metamorphism noted in some ophiolitic rocks seems to be accompanied by a marked decrease in Koenigsberger (Q) ratio to less than 1 (Beske-Diehl and Banerjee, 1980). Low Q ratios are common in samples from both terranes (Table 2), although values somewhat higher than 1 were obtained from the diabase at site 3. Among samples studied by Beske-Diehl and Banerjee (1980) that did not show the decrease in Q was a greenschist pillow with a Q of 31.6 that was found to contain maghemite in addition to low-Ti titanomagnetite. Maghemite has now been reported from several ophiolites (see summary by Luyendyk et al., 1982), but its presence is not always indicated by high Q values. Because pure maghemite is metastable and easily inverts to hematite at low temperatures and pressures, Beske-Diehl and Banerjee (1979) concluded that it could not form in the oceanic crust but must be a result of subaerial weathering. Moreover, dike rocks from the Othris ophiolite (east-central Greece) that contain maghemite were found to possess a hard VRM that was stable near the present-day field direction at the site (Beske-Diehl and Banerjee, 1979).

Although the thermomagnetic measurements on the Mount Diablo samples do not indicate the presence of maghemite, subaerial weathering must be at least partly responsible for the hard overprint. As mentioned in a previous section, the direction of the overprint indicates that it was acquired in the geologically recent past. Samples from site 4 that could be magnetically cleaned came from one of the upper levels on the main quarry face where more than 50 m of overburden has been removed. The overprint could not be removed from 9 of 10 samples collected below the ramp to the quarry floor, which is within a few meters of the original erosional surface, even though the rocks are not visibly more weathered than those on the working face. Sites within the Van Arsdale terrane have also acquired some recent viscous components of magnetization but the VRM is obviously not as hard as in the Mount Diablo rocks even though all sites are surface exposures. Part of the difference may be that the exposures at sites 2 and 3 are unusually fresh because of the vigorous downcutting of the Eel River, and site 1 is also a stream exposure.

Tectonic significance

Before attempting to interpret our paleomagnetic data, we make the following assumptions: (1) The magnetization recovered is that which the rocks acquired at, or very near, the time of their formation. The greenschist-facies metamorphism noted for the Mount Diablo ophiolite is presumed to be due to the ubiquitous hydrothermal alteration occurring at spreading centers. (2) The age of the rocks is reasonably well known. Isotopic ages determined for the Mount Diablo and other Coast Range ophiolite remnants establish that those rocks were emplaced between 165 and 153 Ma. Radiolarians recovered from red chert of the Van Arsdale terrane indicate that this ophiolite is Late Jurassic in age. Isotopic evidence that parts of the Van Arsdale terrane ophiolite could be as young as Early Cretaceous is much more equivocal. (3) Structural attitudes that can be used to correct the paleomagnetic data are well determined. Pillow basalts that are shown to be intruded by the dikes at Mount Diablo, and pillows that were sampled and directly overlie diabase in the northern part of the Van Arsdale terrane are excellent indicators of paleohorizontal. A disconformable relationship between Upper Cretaceous sandstone and underlying diabase, pillow basalt, and radiolarian chert near site 1 also provides a reasonable tectonic correction. (4) Geomagnetic secular variation has been averaged to an acceptable degree. Although we cannot demonstrate conclusively that the full range of secular variation was sampled, the direction of the paleofield at our localities is probably correct to within the uncertainty limits given in Table 1.

Accepting, for the moment, that the assumptions above are correct (and they represent the most straightforward manner in which all the available data can be viewed), what conclusions can be reached about possible tectonic disruptions of the two ophiolites? In order to test for possible rotation or translation of either the Mount Diablo ophiolite or the Van Arsdale terrane, our results must be compared with reference poles of the appropriate age from cratonic North America. Unfortunately, the Jurassic was a period of rapid apparent polar wander (APW) due to the breakup of Pangea and the opening of the present-day Atlantic Ocean, and paleomagnetic data that would accurately determine the APW path are still relatively sparse. The APW paths constructed by Irving and Irving (1982), Gordon et al. (1984), and May and Butler (1986) differ considerably for these reasons. Moreover, all three paths rely, to some extent, on paleomagnetic data from the Colorado Plateau which has undergone a small amount of post-Mesozoic rotation relative to the craton (Bryan and Gordon, 1986; Steiner, 1986, 1988). To accurately reflect the present state of uncertainty in the Jurassic APW path, Mankinen et al. (1989) reviewed all data exclusive of those from the Colorado Plateau and produced the approximate path shown in Fig. 7. Dashed sections

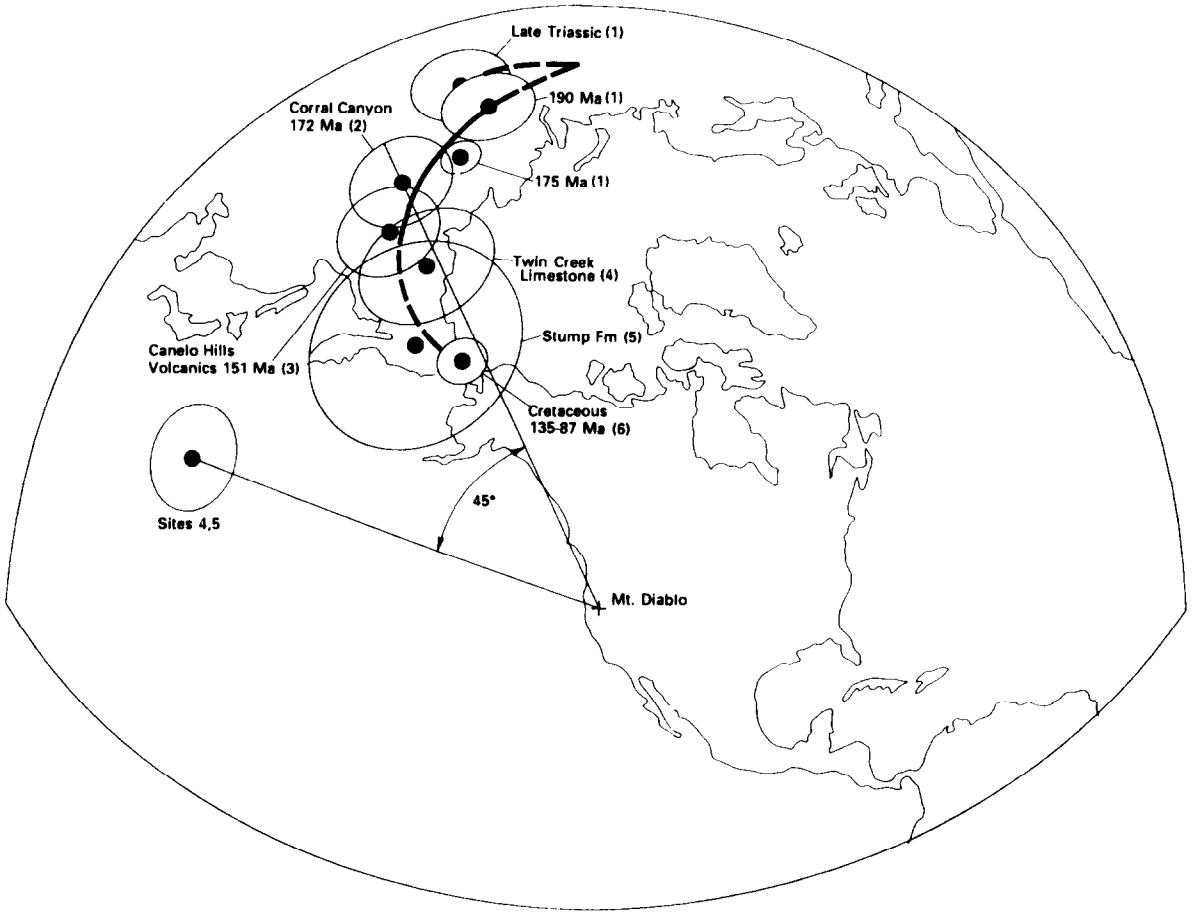


Fig. 7. Schematic map showing mean pole and associated 95% confidence limit for the Mount Diablo ophiolite. Shown for comparison is the Late Triassic to Cretaceous portion of the apparent polar wander path for North America from Mankinen et al. (1989). Path is shown as dashed where poorly determined (see Mankinen et al. (1989) for discussion). References for individual poles: (1) Mankinen et al. (1989), (2) May et al. (1986), (3) Kluth et al. (1982), and May and Butler (1986), (4) McCabe et al. (1982), (5) Schwartz and Van der Voo (1984), (6) Gromme (1984). See Fig. 2B for locations of sites 4 and 5.

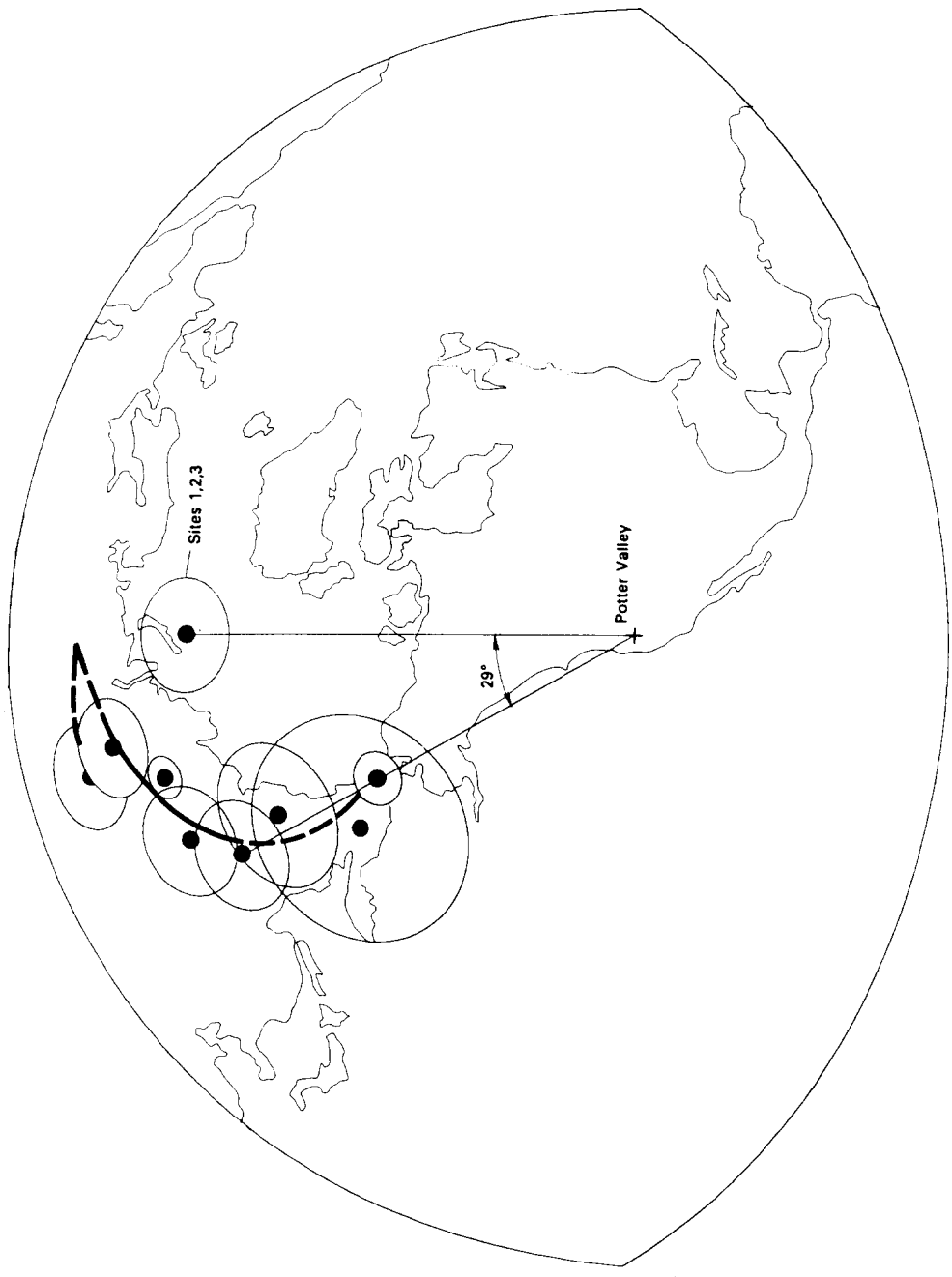


Fig. 8. Schematic map showing mean pole and associated 95% confidence limit for the Van Arsdale terrane. See Fig. 7 for explanation of apparent polar wander path and references for individual poles (shown without labels for clarity). See Fig. 2A for locations of sites 1, 2, and 3.

of Fig. 7 denote time periods during which the path is poorly constrained and, indeed, recent publications (Ekstrand and Butler, 1989; Van Fossen and Kent, 1990; Witte and Kent, 1990) indicate that major revisions may be necessary for Early to Middle Jurassic time. Because the time frame in question is older than the rocks of our study, these latest developments have not been incorporated into Fig. 7.

The paleomagnetic pole calculated for the Mount Diablo ophiolite (Table 1) does not lie along any part of the APW path (Fig. 7), indicating considerable rotation relative to stable North America since the dikes were emplaced. Comparison of the Mount Diablo results with the paleopole calculated for the 172-Ma ash-flow tuffs of Corral Canyon (May et al., 1986) yields a counterclockwise rotation of $45° \pm 7°$. Uncertainties given are 95% confidence limits calculated by the method of Demarest (1983). After correction for tectonic rotation, the Mount Diablo pole is only slightly displaced along the APW path ($6.5° \pm 6.5°$ in latitude) from the Corral Canyon pole. The discrepancy between the two poles is in the correct sense (toward the 151-Ma Canelo Hills Volcanics paleopole), considering that the diabase at Mount Diablo is younger than the ash-flow tuffs of Corral Canyon. Therefore, no latitudinal displacement is indicated for the Mount Diablo ophiolite.

The paleomagnetic pole determined for the Van Arsdale terrane also is discordant (Fig. 8). The youngest reliable Late Jurassic pole with which the Van Arsdale terrane paleopole can be compared is that determined for the Canelo Hills Volcanics (Kluth et al., 1982; May and Butler, 1986). Comparison of these two poles indicates a rotation of $29° \pm 8°$ clockwise and no latitudinal displacement ($0° \pm 7°$) for the Van Arsdale terrane. However, we realize that this comparison may not be entirely appropriate because of the lack of absolute ages for the ophiolite at this locality. If the rocks are considerably younger than 150 Ma, some northward transport could be implied and could be as much as $7° \pm 5°$ if the ophiolite was as young as the reference sections (ranging from 135 to 87 Ma) that constitute the Cretaceous reference pole. This latter possibility seems very unlikely, however, and we consider that there is, at present, no compelling reason to suspect a large latitudinal displacement for this terrane.

Discussion and conclusions

Coast Range ophiolite

Using our paleomagnetic data to restore the Mount Diablo block to its original position indicates that the original orientation of the dikes (and spreading ridge) was approximately N80° W with respect to stable North America (Williams, 1983b). This finding implies a period of north-northeast convergence that was oblique to the N40° W trend of the Jurassic Sierra Nevada magmatic arc. Because of the lack of an age progression of ophiolite remnants either northward or southward, Hopson et al. (1981) suggested that the Coast Range ophiolite represents a spreading ridge which must have been approximately parallel to the present Coast Range trend. Although the original orientation of the Mount Diablo dikes is oblique to both trends, the discrepancy ($\approx 40°$) is small enough that an age progression probably would not be apparent. Because of the north-oblique convergence along the western edge of North America since Late Jurassic time (Engebretson et al., 1985), however, the spreading center represented by the Mount Diablo dike swarm must have been close to the mainland in order to explain the lack of latitudinal displacement. Clockwise rotation of discrete blocks would be expected from such convergence and, in fact, is typically encountered in paleomagnetic studies along western North America (see summary by Beck, 1980). In the case of the Mount Diablo ophiolite, however, the counterclockwise rotation observed is most likely a consequence of development of the piercement structure rather than the result of the direction of convergence or dextral shear along the western margin of the North American plate.

North of Mount Diablo, the Stonyford and Paskenta ophiolites (Fig. 9) were sampled by Frei and Blake (1987), who showed that both were remagnetized, probably during the Cretaceous. However, available paleomagnetic data from the Great Valley sequence in northern California show

no evidence of significant latitudinal displacement (Irwin et al., 1984), a finding that should apply also to its basement rocks which constitute the Coast Range ophiolite. The Coast Range ophiolite is delineated at the western boundary of the Great Valley by a high-amplitude aeromagnetic anomaly

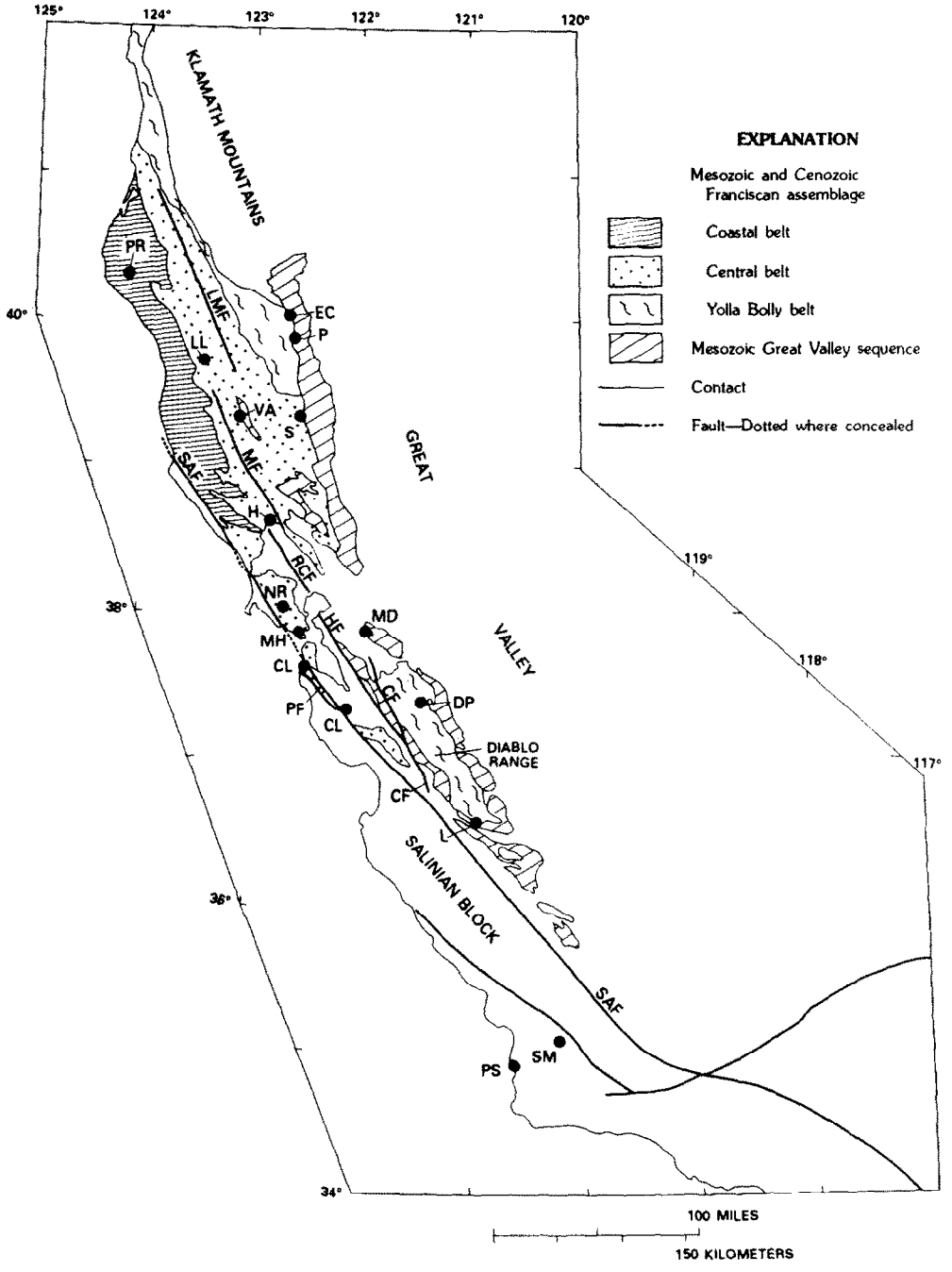


Fig. 9. Map showing localities and units referred to in the text. Geology of the northern Coast Ranges generalized from several sources. *CL* = Calera Limestone; *DP* = Del Puerto ophiolite; *EC* = Elder Creek; *H* = Healdsburg; *L* = Llanada; *LL* = Laytonville Limestone; *MD* = Mount Diablo; *MH* = Marin Headlands; *NR* = Nicasio Reservoir; *P* = Paskenta; *PR* = Parkhurst Ridge; *PS* = Point Sal; *S* = Stonyford; *SM* = Stanley Mountain; *VA* = Van Arsdale terrane. *CF* = Calaveras fault; *HF* = Hayward fault; *LMF* = Lake Mountain fault; *MF* = Maacama fault; *PF* = Pilarcitos fault; *RCF* = Rodgers Creek fault; *SAF* = San Andreas fault.

(Coast Range anomaly), which is continuous from its northernmost occurrence southward to the Diablo Range (Blake et al., 1977). In light of this continuity, and the continuity of the Great Valley sequence rocks (e.g., Jennings, 1977), a finding of no significant translation for the Mount Diablo ophiolite seems entirely consistent with the lack of evidence for translation of the Great Valley rocks in northern California. Numerous outliers of the Coast Range ophiolite and Great Valley sequence have been incorporated into the Franciscan assemblage (McLaughlin et al., 1988), and these must have different displacement histories than those sequences occurring along the western edge of the Great Valley province. Displacements within the northeastern outlier zone of McLaughlin et al., however, are estimated to range from 75 to 260 km and are too small to be detected by the paleomagnetic method.

In contrast to our interpretations, McLaughlin et al. (1988) concluded that the Coast Range ophiolite formed at equatorial latitudes, was transported northward, and was accreted to North America prior to deposition of the overlying Great Valley sequence. This unqualified conclusion is not well founded, however, because the latitude of formation was inferred from preliminary paleontologic and paleomagnetic data reported in abstract form by Hopson et al. (1986). The findings of Hopson et al. refer to the ophiolite remnants at Point Sal, Stanley Mountain, and Llanada (Fig. 9), and the only paleomagnetic results published thus far are from Stanley Mountain (McWilliams and Howell, 1982). Moreover, the ophiolite remnants at Stanley Mountain and Point Sal are parts of the Stanley Mountain terrane of Silberling et al. (1984) which is outboard of the Salinian block (Fig. 9), parts of which have been shown to be far-travelled (Champion et al., 1984). Results from those localities, therefore, are not necessarily applicable to the continuous portion of the unit in central and northern California.

Paleomagnetic data from pillow basalt at Stanley Mountain indicate an origin at a paleolatitude of $14° \pm 7°$ North or South (McWilliams and Howell, 1982), which is not latitudinally discordant if these rocks originated in the northern hemisphere. The expected paleolatitudes for Stanley Mountain are $16.9° \pm 6.2°$ and $23.1° \pm 6.3°$ when the 172-Ma Corral Canyon and 151-Ma Canelo Hills paleopoles, respectively, are used as references. On the other hand, preliminary interpretations of paleomagnetic results from the Point Sal ophiolite ($2° \pm 3°$ North or South paleolatitude) do indicate a significant latitudinal displacement for the Stanley Mountain terrane (Kempner et al., 1976; Beebe and Luyendyk, 1983; Hopson et al., 1986). Preliminary results from a recent reinvestigation of the Stanley Mountain rocks (Hagstrum, 1990b) indicate that the entire sequence of pillow basalt and overlying sediment has been remagnetized, and that some northward transport may have subsequently occurred.

The Llanada ophiolite remnant, which is in the southern Diablo Range approximately 165 km southeast of Mount Diablo (Fig. 9), is part of the Del Puerto terrane. Preliminary interpretation of paleomagnetic results from pillow basalt at this locality (Beebe and Luyendyk, 1983) indicates apparent rotations of $110° \pm 17°$ counterclockwise (assuming a normal polarity for the data) or $70° \pm 17°$ clockwise (assuming a reversed polarity), and an equatorial paleolatitude ($2° \pm 2°$ North or South). Although published abstracts concerning Point Sal, Llanada and, most recently, Stanley Mountain are suggestive of an exotic origin, these results cannot be evaluated critically until the data are made available.

North of Llanada and approximately 75 km southeast of Mount Diablo in the northern Diablo Range, Saad (1969) obtained paleomagnetic samples from serpentinized ultramafic rocks of the Del Puerto ophiolite (Fig. 9) and found that those rocks possess a chemical remanent magnetization (CRM) acquired during serpentinization. Because the attitude of the ultramafic rocks at the time they acquired their CRM is not known, no tectonic significance can be placed on Saad's (1969) results. Tuffaceous chert overlying the volcanic member of the Del Puerto ophiolite has been shown to be remagnetized (Frei and Blake, 1987) and is similarly unsuitable for providing details concerning the tectonic history of this ophiolite.

In summary, no paleomagnetic data are currently available to confirm an exotic origin for the Coast Range ophiolite. If future work does show a

latitude anomaly for some of the south-central California remnants, as seems likely, the boundary between those Coast Range ophiolite remnants that apparently formed close to their present latitude with respect to stable North America and those that may have originated far to the south of their present location must be south of Mount Diablo. The boundary may be found to lie within the Diablo Range if the preliminary results from the Llanada ophiolite prove to be correct. Such a finding would require reassignment of the Mount Diablo ophiolite from the Del Puerto terrane to the Elder Creek terrane of Silberling et al. (1984, 1987).

Franciscan assemblage

The first paleomagnetic data to be reported from rocks of the Franciscan assemblage were obtained from diabase and pillow basalts exposed in the Central belt of Franciscan rocks north of San Francisco (*NR*, Fig. 9). Gromme and Gluskoter (1965) showed that these rocks possess a stable magnetization that yields a paleomagnetic pole in the mid-Atlantic Ocean, far removed from other Late Jurassic and Cretaceous paleomagnetic poles that had been previously reported. Gromme (1984) re-evaluated the earlier data in light of new geologic information and paleontologic age determinations that showed the rocks to be Valanginian in age. He reported a clockwise rotation of $106° \pm 9°$, and a northward translation of $20° \pm 8°$ for what is now referred to as the Nicasio Reservoir terrane by Silberling et al. (1984).

Approximately 30 km south of the Nicasio Reservoir terrane locality (Fig. 9), Curry et al. (1984) obtained paleomagnetic samples from Lower and Middle Jurassic pillow basalt and overlying chert, and from overlying Cenomanian clastic sedimentary rocks of the Marin Headlands terrane (Silberling et al., 1984). Because a primary magnetization for these rocks could not be demonstrated, no conclusion about the latitude of origin was possible. However, all the rocks were obviously magnetized before rotation of the terrane, and Curry et al. (1984) concluded that the Marin Headlands had rotated clockwise nearly 130° since the Cenomanian.

Paleomagnetic results from Franciscan rocks, other than ophiolites, have been reported from the limestones at Laytonville, Calera, and Parkhurst Ridge. The first of these were from blocks of pink pelagic limestone enclosed in melange at the western edge of the Central-belt Franciscan north of Laytonville (*LL*, Fig. 9). Biostratigraphic study of the limestone enabled Alvarez et al. (1980) to determine both the age and stratigraphic top of these exotic blocks and allowed them to conclude that the observed paleomagnetic inclinations implied deposition at $17° \pm 7°$ South paleolatitude. A southern hemisphere origin was reaffirmed by Tarduno et al. (1986, 1990). These blocks were incorporated into the melange by the end of the Cretaceous; thus, extremely fast convergence rates (38 cm/yr) between the Farallon and North American plates are suggested.

The Calera Limestone, which is roughly coeval with the Laytonville Limestone, was sampled at two quarries south of San Franciso (*CL*, Fig. 9) by Courtillot et al. (1985). Biostratigraphic and paleomagnetic study of this limestone indicated deposition at $24° \pm 4°$ North paleolatitude. Results from additional sampling at one of the quarries by Tarduno et al. (1985) indicate that the paleolatitude of the site ranged from 18°N at 105 Ma to 25°N at 90 Ma.

The third occurrence of limestone to be considered is that exposed on Parkhurst Ridge (*PR*, Fig. 9) in melange of the Coastal-belt Franciscan. Preliminary paleomagnetic results by Harbert et al. (1984) indicate that this limestone was deposited at a paleolatitude of 28°, probably in the northern hemisphere. Provided that additional experiments confirm these early results, it appears that at least part of the Coastal belt of the Franciscan has undergone northward transport of a magnitude similar to that of the Calera Limestone but less than that of the Laytonville Limestone.

It is intriguing that the paleolatitudes reported for the Calera and Parkhurst Ridge Limestones and that which can be calculated for the Nicasio Reservoir rocks ($28°N \pm 7°$) all indicate essentially the same amount of northward displacement. Is it possible that these widely spaced localities yield a similar paleolatitude through coincidence or could they have all been part of the same

block? We note that all the localities are contained within what Herd (1978) termed the "Humboldt plate". Although Herd considers the Humboldt to be a newly forming plate because of the recency of faulting along its eastern margin and the fact that these fault zones have not yet coalesced into a single surficial break, is it possible that some ancient fault is being reactivated? The Laytonville Limestone results are anomalous but these blocks are very close to the eastern edge of the proposed plate and may have been caught between it and undisplaced terranes to the east. Of course these ideas are only speculation because of the paucity of reliable paleomagnetic data, but they can be tested in future investigations.

In summary, published paleomagnetic studies of the Coast Range ophiolite and the Franciscan assemblage are scarce because of the difficulties in recovering the primary magnetization of those rocks. Problems in interpretation also arise because the formations are broken, so little continuity exists between outcrops and continuous sequences are difficult to find. Recently, Hagstrum (1990a) reevaluated paleomagnetic data from some of the Franciscan assemblage rocks and considered many of them to have been remagnetized during accretion. Hagstrum showed that some latitudinal transport of individual blocks had occurred after the subducting slab on which they were riding was disrupted, but noted that the amounts indicated by this scenario are compatible with displacements predicted by modern analogs and current plate reconstruction models. Thus, our discussion above and Hagstrum's (1990a) analysis both indicate that paleomagnetic information pertaining to the tectonic development of western California remains largely unsatisfactory. Our limited data from the two ophiolite sequences does little to resolve the outstanding problems, but does raise two important points. First, evidence to support the contention that the Coast Range ophiolite originated at equatorial latitudes and was subsequently transported northward prior to accretion is largely lacking. Paleomagnetic studies seem to indicate that at least some of the terranes southwest of the present-day San Andreas fault are far-travelled, and it would, therefore, be reasonable to expect that outliers of the Coast Range ophiolite in southwest California and in Baja California have also undergone large northward displacements. (For an alternative viewpoint, see the recent abstract by Butler et al., 1990). Our data from Mount Diablo, however, is suggestive that the continuous part of the Coast Range ophiolite that is delineated by the Coast Range aeromagnetic anomaly probably originated close to the continental margin, and that there may be a boundary between nearly autochthonous and truly exotic ophiolite remnants south of Mount Diablo. Second, the fact that our data from the Van Arsdale terrane conflicts with other paleomagnetic results from the Franciscan assemblage may indicate that ophiolitic rocks and limestone knockers in the Franciscan assemblage were not carried to California at the same time as part of a single coherent oceanic plate but are, most likely, the dismembered scraps from a number of colliding plates. The boundary between blocks that formed close to the mainland and those that have been transported significant distances is poorly constrained but must be west of Mount Diablo and Potter Valley. One possibility is that the boundary coincides with the eastern edge of the Humboldt plate (Herd, 1978) that is roughly defined by the Hayward-Rodgers Creek, Maacama, and Lake Mountain fault zones.

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